Geistlinger 2020 HGSOC — analysis summary

RUN 1 OF 2

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

5 patients share ONE continuous manifold (single kNN component; iLISI 1.87/5, limited by 54% T59 dominance). θ=5 resolves a T59-only under-integration lobe seen at θ=2. Judged on the embedding.

Gate 2 — module geography → GO (non-branching)

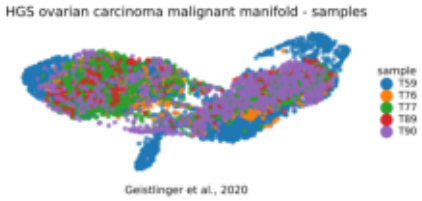
Embryonic pattern present, non-branching: pSom↔Crest single inverse axis (r=−0.95 smoothed); Gast co-varies with pSom (+0.61), both oppose Crest. Crest is the terminal gradient (crest-high decile at pseudotime 0.98). Not a proliferation artifact — pSom↔Crest holds in non-cycling cells (r=−0.62). Crest at the terminus, not the Gast arm — no flag.

Gate 3 — hub cross-repression → NA

Branching-only. No bifurcation (fold-free 2-tip curve; PPT branch-node counts unstable 3–8, no reproducible fork; Gast⊥pSom, r≈0) — no hub cross-repression test. Non-branching case, per §1–2.

STEP FIGURES

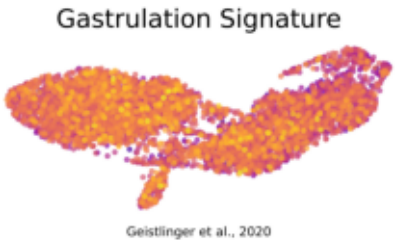
Manifold — samples



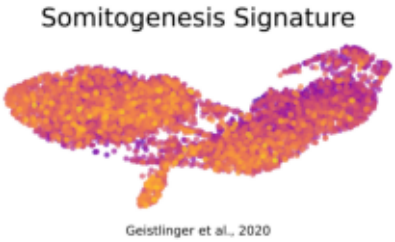
Pseudotime



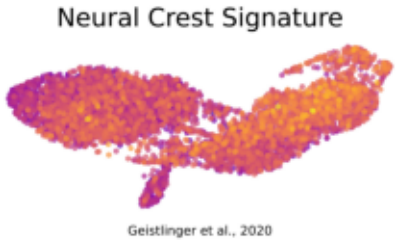
Gastrulation



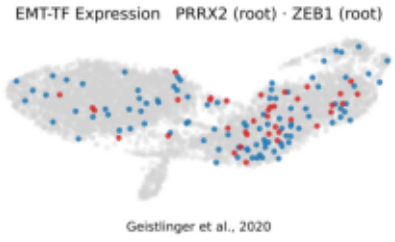
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
High-Grade Serous Ovarian Carcinoma (Geistlinger et al., 2020), 10x — 5,914 malignant cells (5 patients). Harmony integration on sample (θ=5), principal-curve trajectory (scFates) on the whole manifold.

Verdict.
The embryonic-program pattern REPRODUCES as the non-branching case: one continuous gastrulation/somitogenesis → neural-crest developmental axis, rooted at the Gast/pSom-high origin, with Crest an inverse terminal gradient — a strict linear subset of the Puram branching hub. Gate 1 GO, Gate 2 GO (non-branching), Gate 3 NA. No hub / no cross-repression (branching-only).

Parameter choices

Min-cell threshold	≥100 floor → 5/5 kept (gap at head 3180→1298; precedent floor)	**n_pcs / integration**	20 PCs · Harmony θ=5 (merges T59 under-integration lobe)
Island removal	radius-CC, 17-cell speck (740-cell θ=2 lobe was under-integration, kept)	**Geometry**	Non-branching · scFates curve (fold-free 2-tip, self-approach 2.97)
Root	Gast/pSom-high tip (Crest-low); Crest developmentally late	**Subtype annotation**	TCGA + 3CA MPs + Leiden (~85% DIF, continuous gradient)
EMT-TFs shown	PRRX2 · ZEB1 (both root/mesenchymal-enriched, opposite Crest)		